Sample Course Guide for Naive RAG

This is a practice document for testing a PDF question-answering application. All course details below are fictional sample data.

1. Python Foundations

• Duration: 4 weeks.

• Topics: variables, data types, conditions, loops, functions, lists, dictionaries, and file handling.

• Mini project: Student Marks Calculator.

• Prerequisites: No prior programming experience is required.

2. Machine Learning

• Duration: 3 weeks.

• Topics: data preparation, training and testing datasets, linear regression, classification, and model evaluation.

• Mini project: House Price Prediction using sample data.

• Prerequisites: Basic Python knowledge.

3. Generative AI and LLMs

• Duration: 4 weeks.

• Topics: tokens, embeddings, transformers, prompts, and large language models (LLMs).

• Mini project: A simple question-answering chatbot.

• Prerequisites: Basic Python knowledge.

4. Retrieval-Augmented Generation (RAG)

• Duration: 2 weeks.

• Topics: PDF text extraction, chunking, embeddings, vector search, retrieval, and answer generation.

• Mini project: Upload a PDF and ask questions about its contents.

• In this sample guide, FAISS is used for vector similarity search.

5. Assessment and Completion

• Assessment: One quiz, one coding exercise, and one mini project.

• Completion requirement: Submit the mini project and explain how it works.

• Support: Practice questions should be based on the course material.

6. Frequently Asked Questions

• Question: How long is the RAG module? Answer: 2 weeks.

• Question: Which tool is used for vector similarity search in the sample project? Answer: FAISS.

• Question: What is the RAG mini project? Answer: Upload a PDF and ask questions about its contents.

• Question: What is required before Machine Learning? Answer: Basic Python knowledge.

Testing tip: After uploading this PDF, try asking: 'How long is the RAG module?', 'What topics are covered in Generative AI?', and 'Which tool is used for vector similarity search?'